Innovative Infrastructure Solutions (2024) 9:449
https://doi.org/10.1007/s41062-024-01720-1

TECHNICAL PAPER

Evaluation of the engineering properties and sustainability of solid masonry blocks produced with recycled concrete aggregates

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Received: 12 July 2024 / Accepted: 15 September 2024 / Published online: 2 November 2024
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Abstract

The rapid expansion of global infrastructure has led to significant natural resource shortages and increasing environmental concerns related to construction and demolition (C&D) waste. Recycling C&D waste presents a viable solution to alleviate resource depletion and promote sustainability. This study explores the use of recycled concrete aggregates (RCAs) derived from C&D waste in producing solid masonry blocks (SMBs). RCAs were employed to replace both natural coarse and fine aggregates in these blocks. Physical and mechanical properties of SMBs were tested, including hardened density, compressive and flexural strengths, and water absorption. The findings showed that decreased strength and increased water absorption for cement-to-aggregate ratio (C: A) increased from 1:6 to 1:24; however, all produced SMBs satisfied the requirements specified by Indian and ASTM standards. The study concluded that using 100% RCAs in SMBs is structurally sound and environmentally beneficial, meeting international standards. Employing recycled aggregate concrete blocks lowers environmental impact and aids in sustainable development by facilitating the life-cycle closure of the building materials.

Keywords Construction and demolition waste · Recycled aggregates · Recycled concrete aggregates · Recycled aggregate concrete · Solid masonry blocks

Introduction

In India, a substantial amount of construction waste is generated daily, constituting a significant portion of the solid waste in landfills. The construction waste typically includes concrete debris, bricks, tiles, sand, dust, wood, metals, paper, plastic, and cardboard. The disposal of this waste has emerged as a severe ecological and social issue in the country. The extensive use of concrete in buildings has played a significant role in advancing global urbanization. Nevertheless, the substantial need for natural aggregates (NAs) and the manufacturing of cement have resulted in ecological issues and the exhaustion of resources [1–8]. Recently, there has been a notable increase in adopting eco-friendly construction methods, aiming to minimize carbon emissions and preserve natural resources. To date, the majority amount of recycled C&D waste has been utilized as a sub-base or base material for road building ("downcycling"), as well as for engineering fill or landfill engineering [9–11]. Only a small fraction is reused in the concrete industry as recycled aggregates, substituting NAs, which is considered a high-value application [12, 13]. Recycling waste concrete materials to produce recycled aggregate concrete (RAC) has emerged as a favourable strategy for attaining sustainable goals [14, 15].

RAC refers to substituting NAs with RCAs from demolished structures. By incorporating RCAs into concrete blends, the construction industry takes a positive stride in repurposing waste, mitigating ecological repercussions, and easing the demand for naturally occurring aggregate reserves [16]. However, it is essential to note that the mechanical strength properties of RAC may encounter challenges due to weaker particles or adhered mortar on aggregate surfaces and potential impurities within the recycled aggregate [17, 18]. Further, the characteristic strength of RAC was significantly influenced by the quality and ratio of RCAs, especially in the finer particle range. Fine recycled

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concrete aggregates (FRCAs) encompass particles smaller than 4.75 mm, frequently exhibiting elevated contaminant levels, less robust particles, and old mortar adhered to their surfaces [19, 20]. These aspects can impact the produced RAC's mechanical properties and durability resilience. As a result, effective production and incorporation of FRCA is of utmost importance in fully unlocking the advantageous potential of RAC in sustainable construction applications [21].

Environmental pollution and energy waste are growing concerns [22, 23]. In India, fired clay brick, a traditional masonry material, has been utilized for millennia, making India the second-largest producer of burnt clay bricks after China [24]. Research on SMBs has gained significant attention, emphasizing low-carbon living and adopting green building practices. This research aims to avoid using clay and contribute to developing sustainable SMBs. Recently, a substantial amount of research has been carried out on the reliable use of RCAs for manufacturing concrete bricks and blocks [25–27]. Utilizing RCAs to develop mechanized molded bricks and blocks offers several advantages compared to employing RCAs in structural concrete. Automated moulding machines typically produce these blocks and bricks by moulding the concrete mixtures through vibration and compaction. Consequently, the workability of the concrete mixtures, such as their slump, is less critical. This process requires less water for the concrete mixes than RAC used in structural applications, resulting in a lower water-to-cement (w/c) ratio and enhancing the strength of the concrete. The reduced water content in the mixtures minimizes creep and shrinkage in the hardened concrete.

Several studies have examined the features related to the durability and mechanical characteristics of bricks and blocks prepared with recycled aggregates. Poon et al. [26] produced molded RAC bricks and blocks using coarse recycled concrete aggregates (CRCAs) with a maximal size of 10 mm. Laboratory trials were conducted to manufacture $225 \times 105 \times 75$ mm bricks, with natural coarse aggregates (NCAs) replaced by RACs (coarse and fine) at 25%, 50%, 75%, and 100%. The research evaluated the compressive strength (CS), flexural strength (FS), drying shrinkage, and density of the bricks. Findings indicated that up to a 50% substitution level, RACs did not significantly impact the CS of bricks. However, at a 100% substitution level, the CS of RAC bricks decreased from 16.2 MPa to 11.8 MPa, a decrease of 27%. Despite this, the RAC bricks met the BS 6073 [28] requirements for CS (>7 MPa) and FS (>0.65 MPa).

Poon and Lam [29] experimentally investigated the impact of C: A ratios and aggregate types on the performance of paving concrete blocks. They discovered that the CS of the paving blocks decreased as the C: A ratio increased and was closely connected to the crushing strength of the aggregates. The CS increased in direct relation to the strength of the blended aggregate. Additionally, the water absorption of the blocks was strongly connected to the water absorption properties of the aggregate particles. In a subsequent study, Poon et al. [30] examined the characteristics of concrete blocks made with inferior-quality recycled aggregates and found that the soil contents in the fine recycled aggregate negatively affected the properties of the blocks. Specifically, the mechanical strength of the blocks diminished as the content of low-grade fine recycled aggregate increased. As the recycled fine aggregate content increased, the density of the blocks decreased. The findings suggest that low-grade recycled aggregates sourced from construction waste sorting facilities have the potential for use in producing non-structural precast concrete blocks. Guo et al. [31] produced RAC masonry blocks with 75% RCAs and evaluated their durability and mechanical characteristics. Their findings indicated that replacing NCAs with RCAs slightly reduced the CS and durability of the concrete building blocks. However, the concrete block with 75% RCAs still met Chinese standards for drying shrinkage, strength, and freeze-thaw resistance. The compressive properties of masonry prisms made with RAC blocks and traditional mortars, including CS, elastic modulus, stress-strain relationship, and shear performance, are similar to those of standard concrete masonry assemblies. Additionally, the RAC masonry blocks had a lower environmental impact than NAC masonry blocks.

As mentioned earlier, extensive literature exists regarding the feasibility of producing concrete bricks by substituting RCAs, aiming to enhance sustainability in construction. Moreover, there has been a significant exploration into manufacturing concrete products like paving blocks (especially for light traffic) and curbs by partially or fully replacing NCAs with RCAs in the concrete mix, yielding promising outcomes [32–37]. Nevertheless, not much study has been done on the potential of producing load-bearing concrete SMBs using 100% RCAs derived from recycling C&D waste and minimizing cement content by increasing the cement-to-aggregate ratios (C: A). Therefore, the current study focused on developing load-bearing solid masonry concrete blocks to address the research gap and promote extensive recycling of C&D waste for sustainable construction practices. The study replaced 100% of NCAs and natural fine aggregates (NFAs) with CRCAs and FRCAs, respectively. Primarily, the study evaluates how the different C: A mix ratio composites affect concrete's fresh and hardened properties. In addition, it aims to evaluate structural efficiency, carbon emissions, and the cost analysis and energy savings of produced SMBs.

Research significance

The outcomes of this research can potentially bring about a significant shift in using sustainable masonry blocks for construction, aligning with the sustainable goals outlined by the United Nations, especially for a country like India. The proposed materials could help cut down waste treatment costs, reduce the use of new materials, and lower CO_2 emissions. They are also validated as sustainable elements that promote green building practices, thus contributing to sustainability efforts. By incorporating RCAs, the abundant waste materials in India, into SMBs, the aim is to create new avenues for enhancing sustainability and substantially easing the material cost burden for affordable housing in society. Furthermore, this research aligns with and supports sustainable development goals, particularly emphasizing green building initiatives. Through showcasing the structural efficacy of masonry and the application of SMBs, this study may establish guidelines to benefit the lower-middle-income segments of society by providing affordable housing through blocks derived from construction demolition waste RCAs.

Materials and characterization

The current study utilized ordinary Portland cement (OPC), CRCAs, FRCAs and water as the primary components for producing SMBs. The details of each material are explained below.

OPC

The study used an OPC of 43 grade, meeting IS 269:2015 specifications [38], with a specific gravity of 3.15 and 285 m^2/kg fineness. The chemical composition of the OPC obtained through the X-ray fluorescence (XRF) test is shown in Fig. 1.

Aggregates

The RCAs used were procured from the site of a 50-year-old demolished building located in the Kasturba Medical College campus in Manipal, Udupi district, Karnataka state, India. The dismantled concrete was crushed using a jaw crusher machine to yield aggregates in their unsorted and ungraded condition, which underwent mechanical sieving. The particles that passed through a 4.75 mm sieve and retained on 150 microns were considered as FRCAs, as shown in Fig. 2(a), and they were checked for physical properties as per IS 2386:1963 part 3 and obtained results are listed in Table 1. Similarly, particles ranging in size from 25 to 4.75 mm were sorted out as CRCAs shown in Fig. 2(b) and subjected to physical and mechanical tests per the IS 2386:1963 parts 1, 3, and 4, and results are tabulated in Table 1. In addition, both the categories of aggregates were checked for particle size distribution according to IS 383:2016 [39]. Results indicated that all the particles were evenly distributed and showed a well gradation curve, as seen in Figs. 3 and 4.

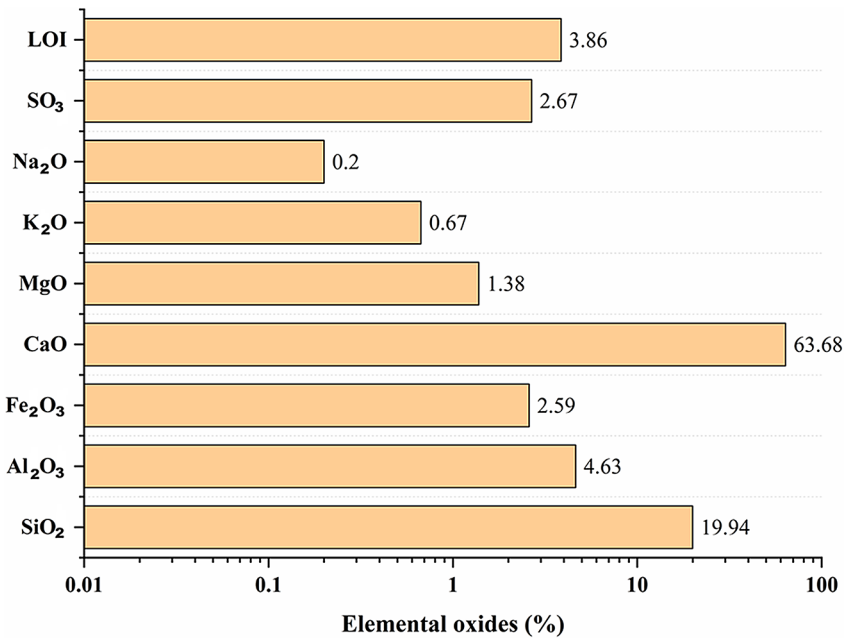


Fig. 1 Chemical composition of OPC



Fig. 2 (**a**) Fine recycled concrete aggregates (**b**) Coarse recycled concrete aggregates

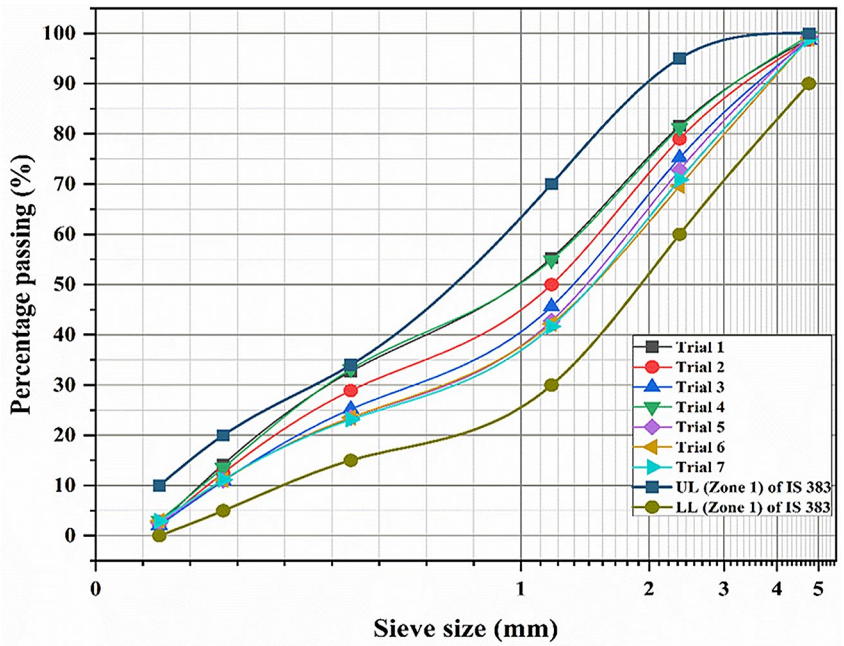
Table 1 Physical and mechanical properties of aggregates

Properties	Referred to	CRCAs	FRCAs
Fineness modulus	IS-383 [39]	7.26	3.22
Bulk density (kg/m^3)	IS-2386 (III) [40]	1342.31	1335.83
Water absorption (%)	IS-2386 (III) [40]	3.95	6.96
Specific gravity	IS-2386 (III) [40]	2.5	2.47
Voids (%)	IS-2386 (III) [40]	46.29	45.92
Impact value (%)	IS-2386 (IV) [41]	34.3	-
Crushing value (%)	IS-2386 (IV) [41]	38.66	-
Abrasion value (%)	IS-2386 (IV) [41]	39.67	-
Flakiness and Elongation Index (%)	IS-2386 (I) [42]	22.89	-

X-ray diffraction (XRD) of OPC and RCAs

XRD patterns of OPC and RCAs are shown in Fig. 5. XRD pattern of OPC indicated the existence of minerals alite (ICSD-4331 & 162744), belite (ICSD-182052), and calcium silicon oxide (ICSD-963) would aid in hydration when exposed to water. The XRD test data obtained for RCA patterns reveals that quartz peaks predominate, with small calcite and anorthite peaks following. The comparatively low height of the calcite and anorthite peaks compared to the prominent quartz peaks indicates their low contents. Calcite may result from the carbonation process within the original concrete during its service life or be directly correlated to the initial cement paste stuck to the surface of RCA particles [43], while quartz presence must be related to adhered mortar on the surface of the aggregate. Similar XRD result patterns for RCAs are demonstrated in research by M.C. Limbachiya et al. [44] and Baifa Zhang et al. [45].

Fig. 3 Gradation curves of FRCAs



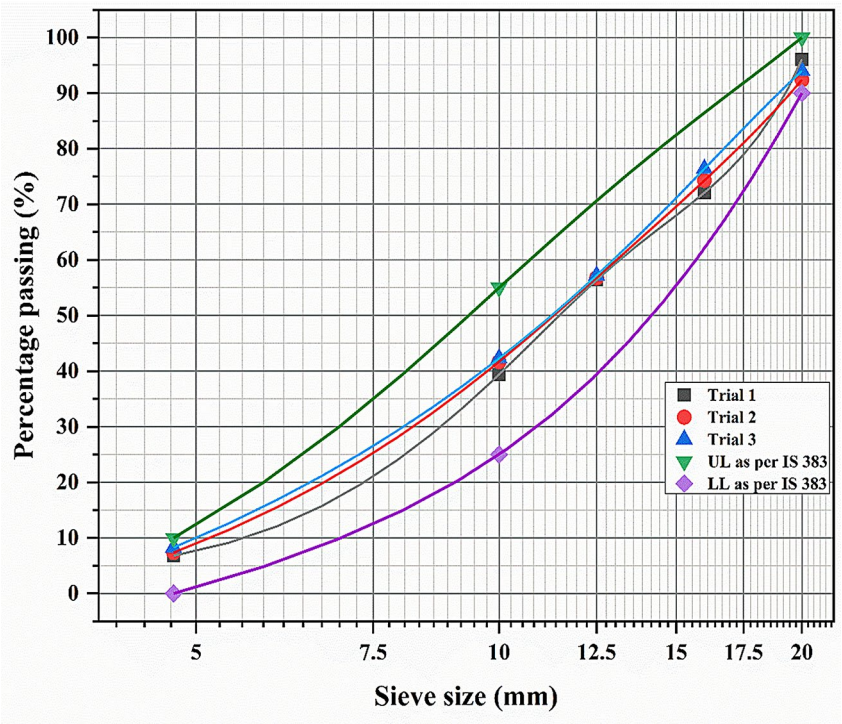


Fig. 4 Gradation curves of CRCAs

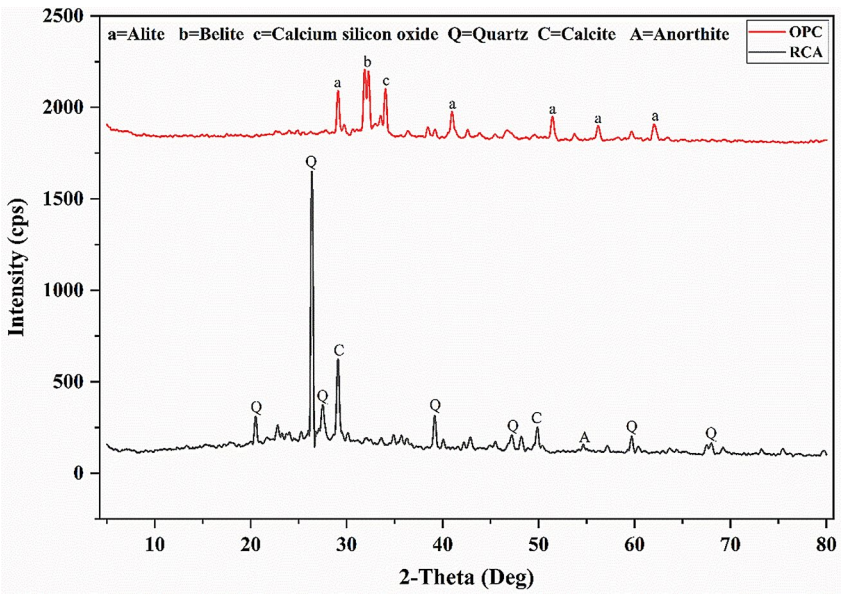


Fig. 5 XRD of OPC and RCAs

Experimentation

Mix design

The SMBs were developed with varying C: A ratios of 1:6, 1:8, 1:11, and 1:13 mix proportions based on findings from previous literature [46, 47]. Further mix proportions were investigated by progressively increasing the C: A ratios to higher levels of 1:15, 1:18, 1:21, and 1:24. This process continued until the mixtures failed to meet the strength requirements specified in IS 2185 (Part 1) – 2005 [46]. These mixed ratios were aimed at making load-bearing SMBs. The desired strength for load-bearing structures differs across ASTM, British, and Indian standards; However, this study adopted the Indian standard due to its higher required threshold and broader usage, targeting a CS of 5 MPa to identify the most suitable combination. The parameters studied included the cement contents and the relative proportion of RCAs in C: A ratios. Eight concrete block mixes were developed, each containing varying amounts of OPC and RCAs. These masonry blocks comprised 100% RCA for coarse and fine aggregates. These mixes are detailed in

Table 2 Mix proportions

Mix Ratio (C: A)	Water (kg/m^3)	OPC (kg/m^3)	FRCAs (kg/m^3)	CRCAs (kg/m^3)	w/c
1:6	153	306	725	1114	0.5
1:8	123	245	774	1189	0.5
1:11	94	189	819	1259	0.5
1:13	82	164	839	1290	0.5
1:15	72	145	855	1313	0.5
1:18	61	123	872	1340	0.5
1:21	53	107	885	1360	0.5
1:24	47	95	895	1375	0.5

Table 2, representing quantities required for producing one cubic meter of concrete blocks. The diverse combinations of C: A ratios are specified by their weights. In this context, 'C' stands for the OPC, 'A' denotes the RCAs, and the number specifies the weight proportions. According to the literature, the effective water-to-cement ratio for all the mixes was fixed at 0.5 [10, 22, 47, 48]. All the mix slumps were maintained between 0 and 10 mm, respectively.

Manufacturing of SMBs

The SMB samples were manufactured in the research laboratory using a dry-mix method outlined in various literature sources [10, 30, 48]. Initially, CRCAs and FRCAs were mixed along with the OPC in a drum mixer for about 3–5 min to attain a uniform mix, followed by adding a calculated quantity of water. Blending continued for another 4–8 min until a homogeneous dry mixture of low workability indicated by a 0 to 10 mm slump value was achieved. As shown in Table 1, recycled aggregates have exhibited higher water absorption, which will lower the overall properties of concrete; hence, the aggregates were saturated in water for one hour before mixing. This one hour of saturation of aggregates is carried out by doing different trials of intervals of 10 min of saturation of RCAs till there was not much effect of water absorption properties of aggregates on the mix. In the dry-mix approach, the pre-soaked water did not affect the water/cement ratio, as the water absorbed by the recycled aggregate helps in the cement hydration process during hardening [49]. The uniform mixture obtained was poured into 400 mm x 200 mm x 200 mm steel mould specified by IS 2185 (part 1)-2005 [46]. The mixture was poured into the mould and compressed using a table vibrator after being layered into three layers. Following compaction, the moulds were left in an undisturbed state for one day and covered with plastic sheets to attain initial strength and kept underneath the dry laboratory conditions in which the temperature has a variation between 25 °C and 30 °C, as specified by IS 456–2000 [50], that the curing should be done under humidity of 90% at a temperature of 27 ± 3 °C. The block specimens were demoulded after one day of casting and kept moist by covering them with wet gunny bags for 28 days to complete the curing process, as shown in Fig. 6.

Tests conducted

Fresh density

The fresh density of SMBs with varying mix ratios was assessed following the guidelines of IS 1199 (part 3)-2018 [51]. Initially, the empty container was weighed to find its mass, m_1, followed by filling it with the concrete mix in two layers with the required compaction to determine its mass, m_2. The container volume (V) was also calculated, and the fresh density was computed using Eq. 1.

$$\rho_{fr} = (m_2 - m_1)/V \quad (1)$$



Fig. 6 Casting and curing of SMBs

Dimensions

The dimensions of the SMBs were determined according to British standard BS EN 771-3:2003 [52], which involved measuring length, width, and height. The variations in dimensions and allowable tolerances were then measured and compared in accordance with the British standards for all the masonry blocks.

Hardened density

The hardened density of the SMBs was assessed following the guidelines of the IS 2185 (part 1)-2005 [46]. The blocks were dried in a vented oven at 100 °C and cooled to room temperature. Their dimensions were measured, and the net volume (V_{net}) was calculated based on length, width, and height. Subsequently, the blocks were weighed to determine their dry mass (M_{dry}) per standards. The hardened density (ρ_b) was then calculated using Eq. 2.

$$\rho_b = M_{dry}/V_{net} \tag{2}$$

Water absorption

The water absorption characteristics of SMBs were evaluated following the IS 2185 (part 1)-2005 [46]. After 28 days, the water absorption was determined by placing oven-dried block specimens in water at ambient temperature for 24 h. The specimens were removed and left to drain on a wire mesh for a minute. Any visible surface water was removed with a dry cloth, and the samples were immediately weighed (M_{wet}). Subsequently, all specimens were dried in a vented oven at 100 °C to 115 °C for 24 h until two consecutive weighings, spaced two hours apart, showed a moisture loss of not more than 0.2% of the previously recorded dry mass (M_{dry}). The water absorption percentage was determined using Eq. 3 as the ratio of absorbed water mass to the oven-dried mass of the same specimen.

$$\text{Water absorption, in } \% = \frac{(\text{Mwet} - \text{Mdry})}{\text{Mdry}} \times 100 \tag{3}$$

Compressive strength

For SMBs, the CS of blocks was assessed following the test methods specified in ASTM C90–09 [53]. These compressive tests were conducted 28 days after the blocks were manufactured. A testing machine determined the CS of blocks with a 3000 kN loading capacity. The load was applied to the masonry blocks of the standard area (400 mm x 200 mm) and increased steadily at five kN/s. For each mix proportion, three specimens were tested, adhering to ASTM C90–09 [53].

Flexural strength

The FS of SMBs was evaluated following the procedure outlined by Dayaratnam, P -1987 [54] and complied with IS 4860–1968 [55] after a 28-day curing period. The testing method involved a three-point bending test with a support span of 300 mm and a height of 200 mm, using a 300 kN maximum load capacity hydraulic universal testing machine. The load was applied gradually to the standard area of the masonry blocks (400 mm x 200 mm) at a rate of 10 kN/min. Each mix proportion was tested with three samples, following the prescribed standards.

Results and discussion

Fresh density

The fresh density is influenced by elements such as the type of aggregates used, the water-to-cement ratio, and voids within the mixture [56]. These factors impact the fresh density of concrete and play a crucial role in determining its hardened properties. A lower density in the fresh state of concrete typically indicates a higher water and void content, which can reduce strength in the hardened state. Generally, RAC exhibits lower density than NA concrete due to low-density adhered mortar [57, 58]. Experimental findings regarding the fresh density of the SMBs for different mixes are depicted in Fig. 7. In mixes, a water-to-cement ratio of 0.5 was used to attain the desired zero to 10 mm slump value for a dry mix of SMB. The fresh density of blocks decreased progressively from a 1:6 ratio to a 1:24 ratio. This reduction in density ranges from approximately 1.18–7.21% compared to the 1:6 mix ratio. The primary reason for the density reduction in higher mix ratios compared to the 1:6 ratio was the increase in aggregate proportions, which led to higher water absorption and more void content.

Geometrical requirements

No variations in the colour and texture homogeneity and uniformity among the samples produced with different SMB mix ratios were noticed. Any change in the colour and texture of the SMBs may affect the masonry blocks' long-term performance and lead to strength loss and aesthetic implications [59]. Table 3 displays the outcomes concerning dimension tolerance and maximum deviations for various mix proportions following BS EN 771–3 [52] standards.

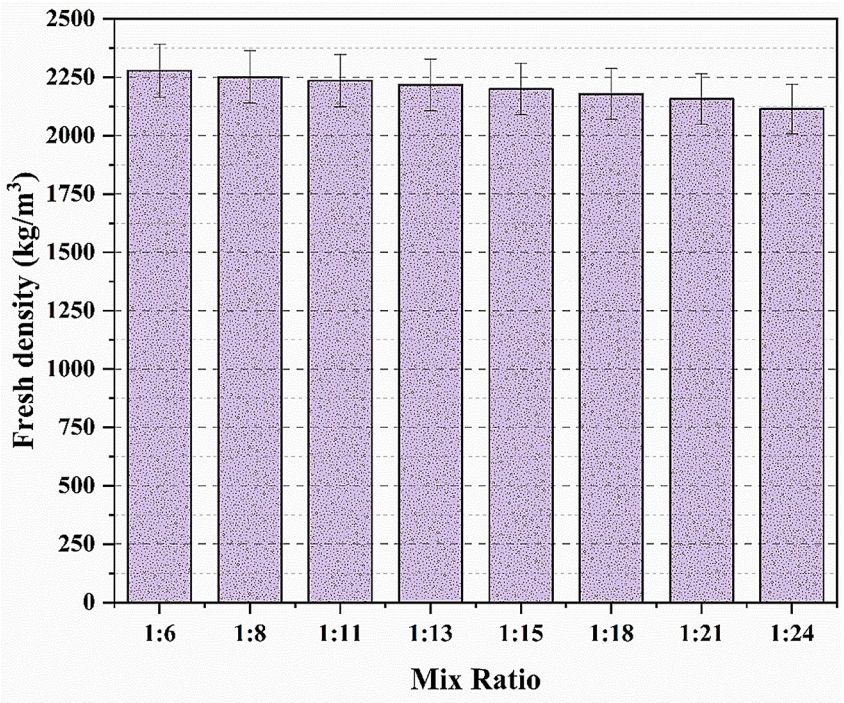


Fig. 7 Fresh density of SMBs for different mixes

Table 3 SMBs limit deviations and tolerance category according to BS EN 771–3 (in millimetres)

Mix Ratio (C: A)	Length	Width	Height	Length deviation	Width deviation	Height deviation	Tolerance Range	Tolerance category
1:6	397.7	198.6	198.4	-2.3	-1.4	-1.6	+1 & -3	D2
1:8	397.5	198.5	198.3	-2.5	-1.5	-1.7	+1 & -3	D2
1:11	397.2	198.3	198	-2.8	-1.7	-2	+1 & -3	D2
1:13	396.9	197.9	197.8	-3.1	-2.1	-2.2	+3 & -5	D1
1:15	396.7	197.8	197.6	-3.3	-2.2	-2.4	+3 & -5	D1
1:18	396.3	197.6	197.3	-3.7	-2.4	-2.7	+3 & -5	D1
1:21	396.2	197.3	197.2	-3.8	-2.7	-2.8	+3 & -5	D1
1:24	395.9	197.1	196.9	-4.1	-2.9	-3.1	+3 & -5	D1

These values represented an average of three measurements. The findings suggested that all block dimensions adhered to specified limit deviations, irrespective of the mix type. No discernible visual distinctions were observed among the different block types. The SMBs utilized in this research were sized at 400 mm × 200 mm × 200 mm, categorizing them under categories D1 and D2 as per BS EN 771-3 [52], as outlined in Table 3. If SMBs do not adhere to dimension tolerance standards, it can impact the structural integrity of the construction. This non-compliance may lead to issues such as improper alignment and stability of the blocks, uneven load distribution across walls, and poor bonding between mortar joints. Additionally, dimension tolerance deviations can affect the structure's thermal and acoustic performance and compromise its aesthetic quality [60]. Figure 8 illustrates the standard-size SMBs commonly utilized in masonry constructions.

Hardened density

The hardened density of SMB for various mix proportions is illustrated in Fig. 9. The values provided represent the average of three separate specimens. The results indicated that the hardened density of blocks decreased as the RCA content increased. The hardened density of blocks decreased for all mix ratios compared to the 1:6 mix ratio, which ranged from 2.10 to 9.96%. The hardened density of SMBs for C: A ratios of 1:6 to 1:24 was 2215.23 kg/m^3 to 1994.59 kg/m^3. In addition, the hardened density of SMBs in each C: A ratio mix has decreased compared to the previous ratio, varying between 0.8% and 2.1%. The density of RCAs significantly influences the density of SMBs. However, it is also affected by the heterogeneity of RCAs, including factors such as particle size distribution, fines content, and the coupling capacity of the aggregates during the vibro-compaction process [48]. Higher mix ratios lead to lower bulk density, lower

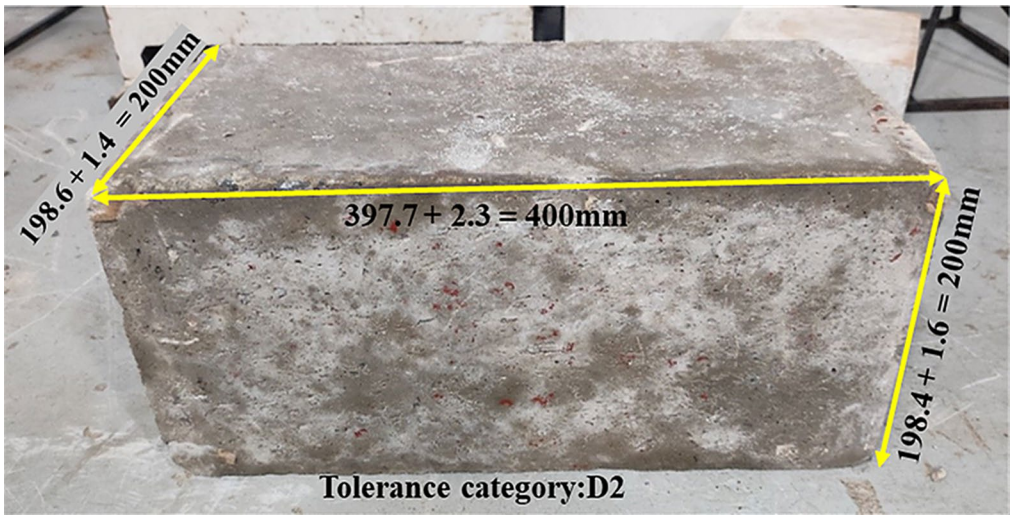
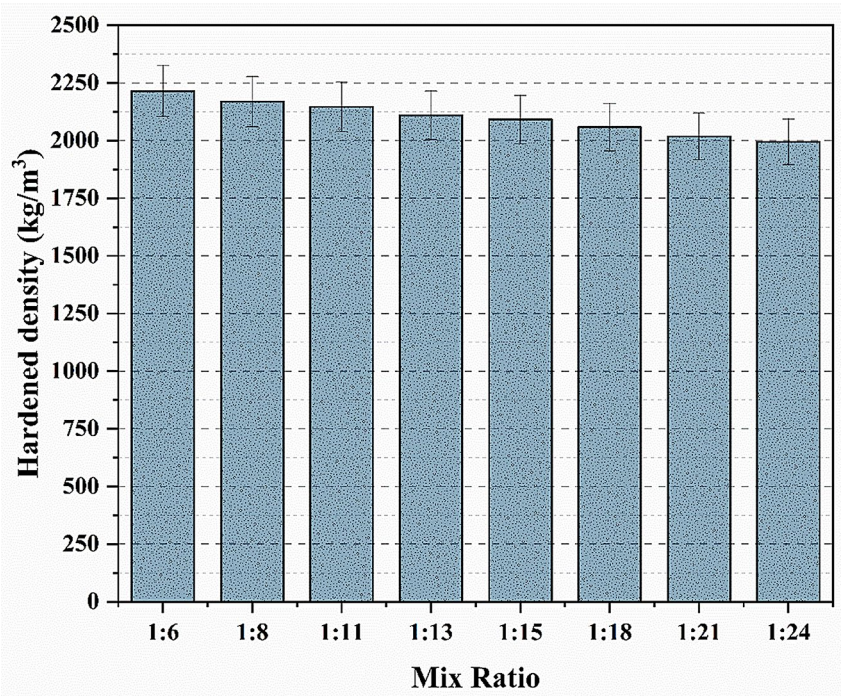


Fig. 8 Dimensions of SMBs

Fig. 9 Hardened density of SMBs for different mixes



specific gravity, and the presence of pores, contributing to a reduction in SMB's hardened density. The results align with the acceptance standards outlined in the IS 2185 (Part 1)-2005 [46] and ASTM C90 [53], which should be greater than 1800 kg/m^3 and 2000 kg/m^3, respectively.

Compressive strength

The CS of SMBs containing various mix ratios of cement and RCAs is illustrated in Fig. 10. It included the dry compressive strength (DCS) and wet compressive strength (WCS) measured at 28 days. For the WCS, the blocks were submerged in potable water for 24 h before being tested in a compression strength machine. Results showed that increased RCA content decreased the DCS and WCS. It implies that a higher ratio of C: A corresponds to a reduction in the strength of the SMBs. For instance, reducing the cement content at a higher C: A ratio reduced DCSs by 13.85–84.51% and WCSs by 16.57–87.93%, respectively, for different ratios compared to a 1:6 mix ratio. This reduction was primarily due to the decreased formation of cement hydration products at higher ratios due to decreased cement content, increased voids, and the porous nature of RCAs. The CS of SMBs for C: A ratios of 1:6 to 1:24 in both DCS

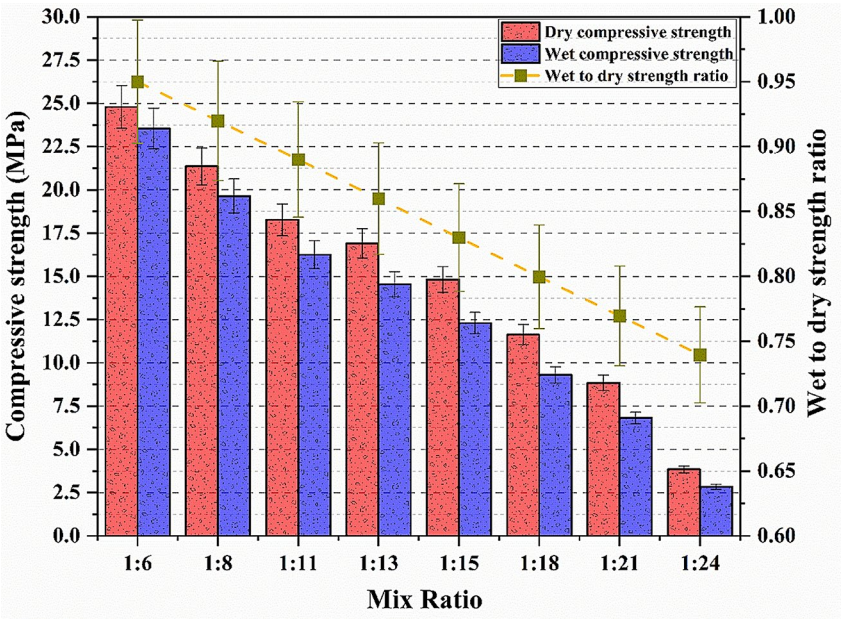


Fig. 10 CS of SMBs for different mixes

and WCS was 24.79 MPa to 2.84 MPa. The CS of SMBs in each C: A ratio mix from 1:6 to 1:15 compared to the previous ratio has decreased, varying between 7.4% and 17.3%; similarly, for C: A ratio mix from 1:18 to 1:24 compared to the previous ratio has also decreased the CS, varying between 21.5% and 58.3%.

Further, the WCSs of the SMBs were inferior compared to their DCSs because the existence of water acts as a lubricant around the permeable pores in the blocks, weakening the resistance between the adjacent bonded particles and making them more susceptible to displacement [61]. The wet-to-dry compressive strength ratio decreased in the range of 0.95 to 0.74 for all the different ratios of SMB, as shown in Fig. 10. The minimal difference between the WCS and DCS in lower C: A mixes like 1:6 and 1:8 was mainly due to the higher cement content compared to higher mixes like 1:21 and 1:24. This increased cement content in a mix positively affects the strength. Hence, the immersion treatment promotes further hydration of the unhydrated cement particles, which helps offset the strength loss caused by the presence of water [62]. The results met the acceptance criteria outlined in the IS 2185 (Part 1)-2005 [46], requiring a CS greater than 5 MPa, which was achieved for C: A ratios of 1:6 to 1:21. Additionally, as per ASTM C90 [53], a CS requirement should be greater than 13.1 MPa, which is satisfied by mix ratios of 1:6 to 1:13 for both wet and dry conditions.

SMB's hardened density is closely linked to their CS. As the density of SMBs increases, so does their CS; on the other hand, a decrease in density leads to a reduction in CS. The density of these blocks is largely influenced by the packing density or packing behaviour of the concrete in the production of SMBs. This packing behaviour creates pores or voids within the concrete, affecting the density and strength of the SMBs. Figure 11 shows a regression analysis demonstrating the strongest correlation between hardened density and CS occurring with a linear function, resulting in a correlation coefficient (R^2) of 0.978.

Flexural strength

The FS results for the blocks, illustrated in Fig. 12, represented the average of three measurements. These values followed a trend similar to that observed in CS. Like the CS development, dry and wet FSs of the SMBs decreased with the increased RCA content. For instance, reducing the cement content at a higher C: A ratio reduced dry FSs by 13.43–80.08% and wet FSs by 17.77–81.07%, respectively, for different ratios compared to a 1:6 mix ratio. The FS of SMBs for C: A ratios of 1:6 to 1:24 in dry and wet conditions was 4.21 MPa to 0.61 MPa. In addition, the FS of SMBs in each C: A ratio mix has decreased compared to the previous ratio, varying between 6.3% and 32.5%. Notably, the FSs of all the block specimens significantly exceeded the requirement of 10% of CS, as specified by Dayaratnam, P -1987 [54], and adhered to the IS 4860–1968 [55].

The block failure patterns were primarily characterized by shear cracks running diagonally in SMB caused by high local tensile stresses within the blocks. Additionally, vertical cracks appeared in the centre of blocks, which subsequently failed due to diverging shear cracks. Figure 13 illustrates the regression analysis, showing that the strongest correlation between CS and FS was achieved with a linear function, resulting in a correlation coefficient (R^2) of 0.89.

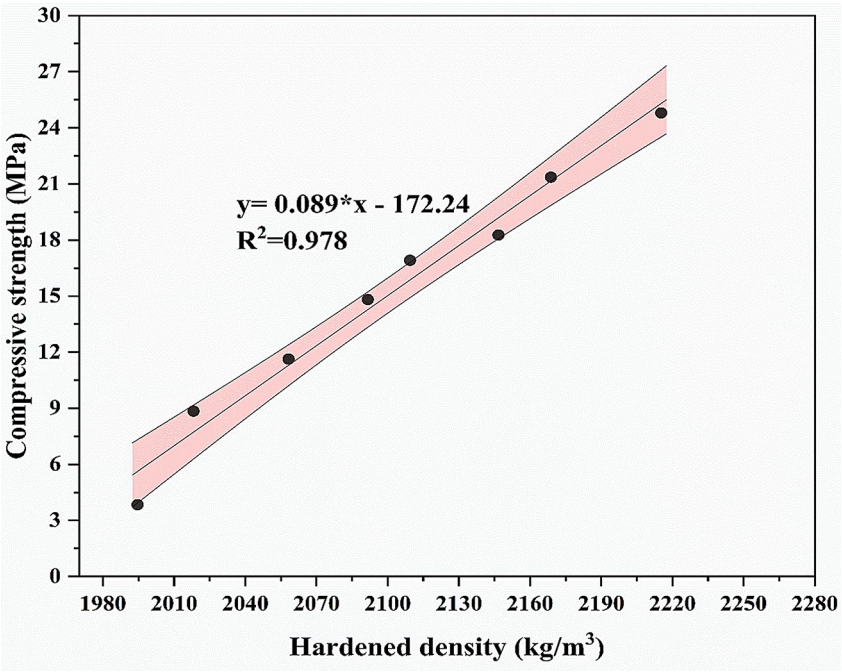


Fig. 11 Correlation between hardened density and CS

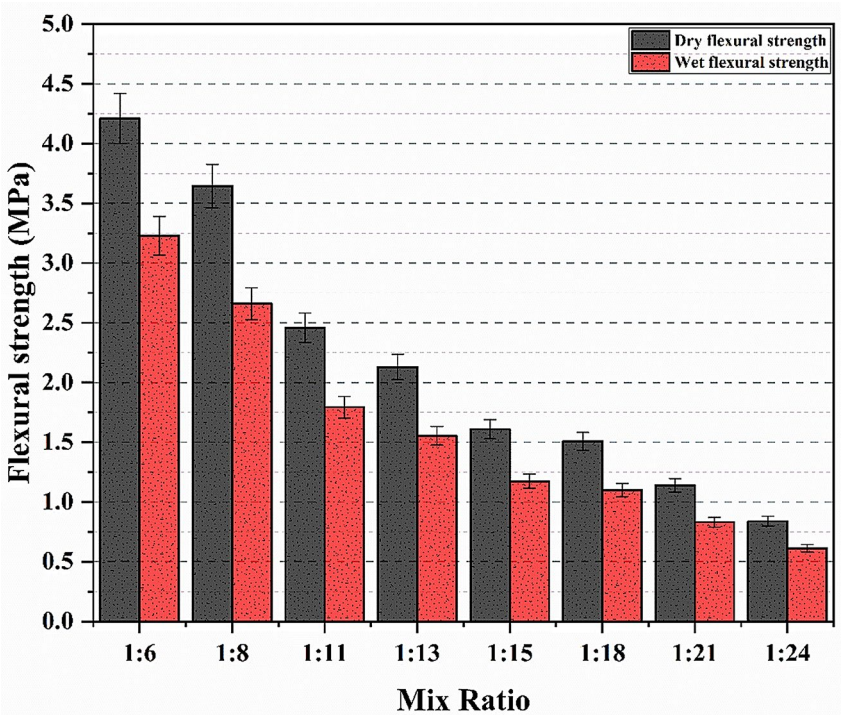


Fig. 12 FS of SMBs for different mixes

Water absorption

Water absorption is a key property in masonry blocks, significantly influencing their durability and the bond strength between the block and mortar in construction. High water absorption can compromise the structural integrity by increasing the porosity and weakening the blocks, reducing overall strength. Conversely, lower water absorption rates indicate higher-quality blocks that are more resistant to weathering and maintain better structural integrity. The quality of the masonry block and the bond with the mortar are both affected by water absorption, impacting the structure's overall strength [63, 64]. For all mix ratios from 1:6 to 1:24, the water absorption values ranged from 1.42 to 3.57%. The percentage water absorption of SMBs in each C: A ratio mix has increased, varying between 5.5% and 29.2% compared to the previous ratio. The increased water

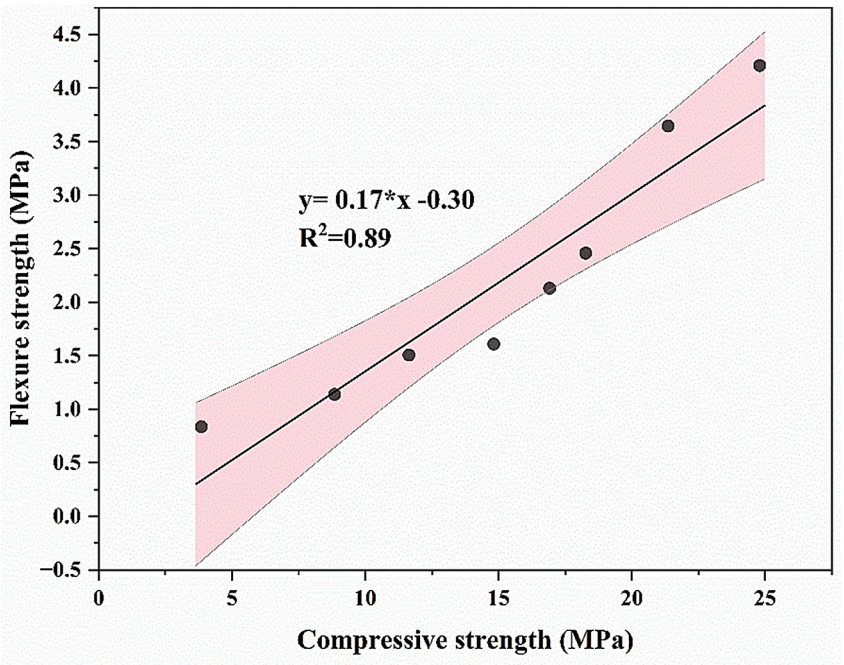


Fig. 13 Correlation between CS and FS

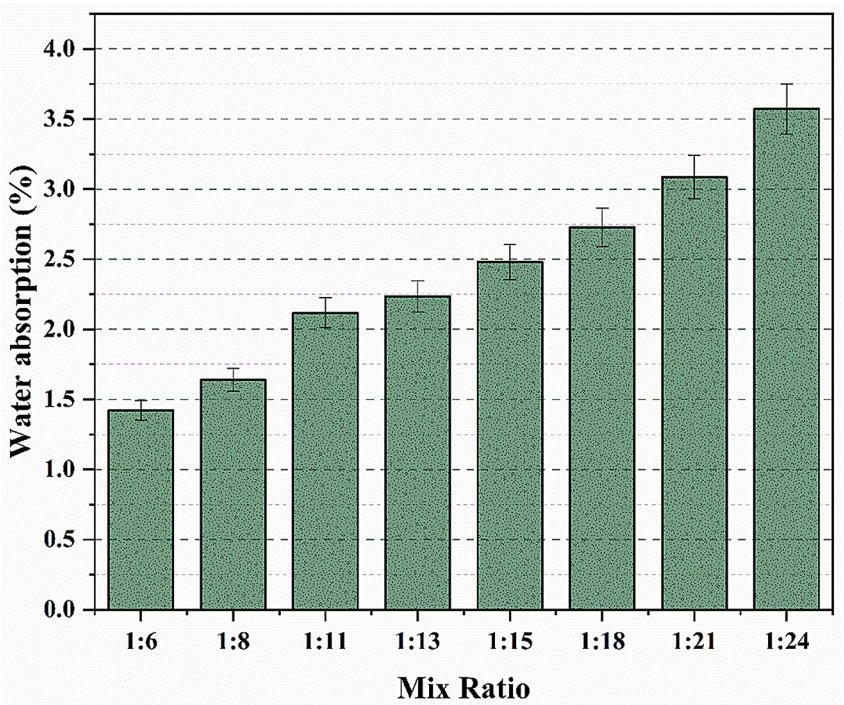


Fig. 14 Water absorption of SMBs for different mixes

absorption with higher mix ratios was primarily due to the increased presence of voids and the porous nature of RCAs. The RCA content was greater than the cement content at higher mix ratios, resulting in a weaker interfacial transition zone and higher water absorption in the SMBs. The water absorption results are depicted in Fig. 14. It is evident that all water absorption values fall below the maximum capacity allowed for SMBs, which is less than 10%, as per IS 2185 (Part 1)-2005 [46]. Since the water absorption results are under permissible limits, there would not be much significant impact on the durability or long-term performance of the SMBs, and there would be no adverse effect on the structural integrity.

The water absorption characteristics of SMBs are mainly attributed to the increased amount of RCAs. Additionally, the reduced cement content and adhered mortar on the aggregate surfaces contribute to higher water absorption in these blocks. Figure 15 presents a regression analysis,

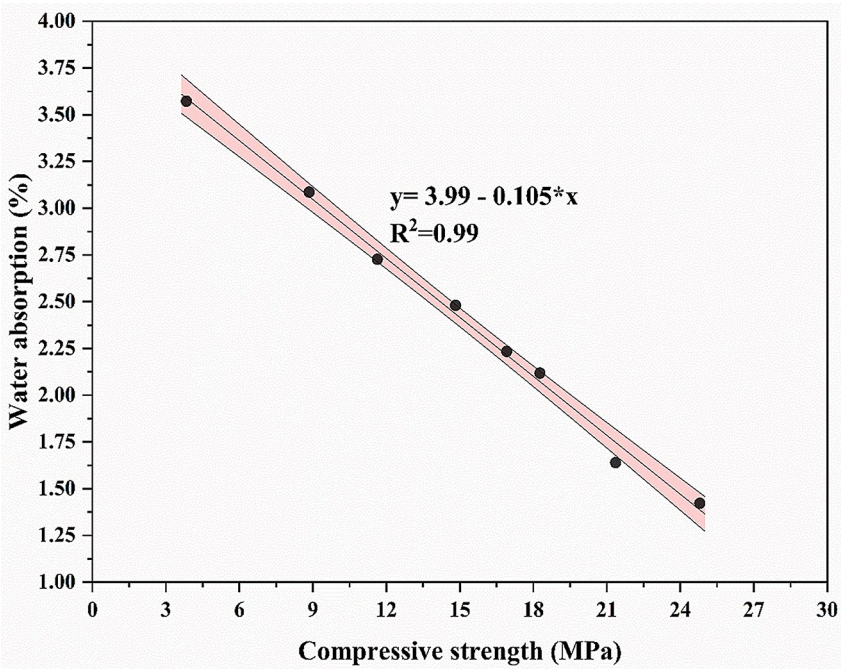


Fig. 15 Correlation between CS and water absorption

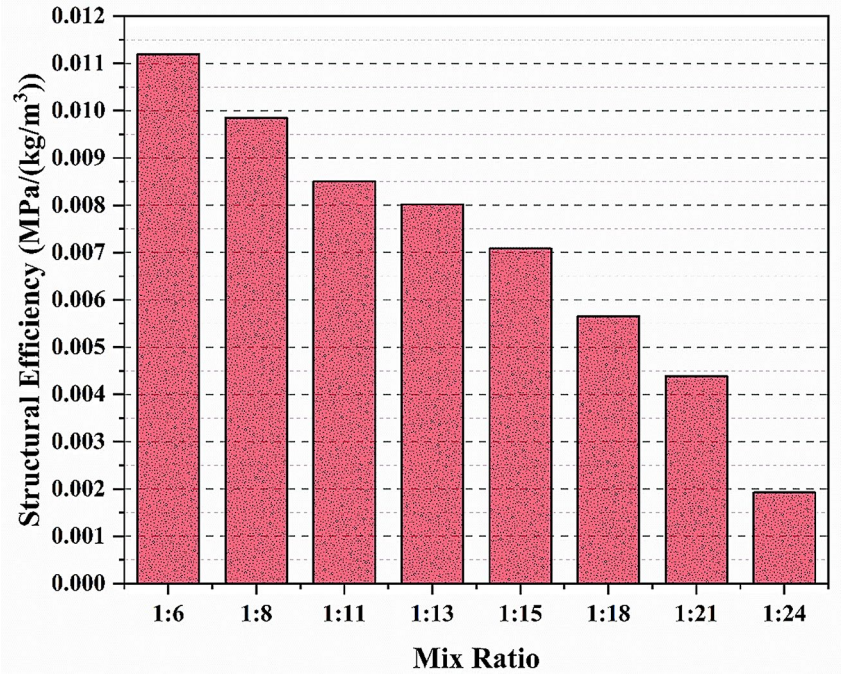


Fig. 16 Structural efficiency of SMBs for different mixes

revealing that the best correlation between CS and water absorption is achieved with a linear function, yielding a correlation coefficient (R^2) of 0.99.

Structural efficiency

Structural efficiency was introduced and assessed by comparing specimens with different mix ratios of RCAs-based SMB [65, 66]. The structural efficiency value, which reflects the relationship between CS and hardened density of SMB after 28 days of curing, is depicted in Fig. 16. The structural efficiency values ranged from 0.0112 in a lower mix ratio of 1:6 to 0.0019 in a higher mix ratio of 1:24. This indicates that structural efficiency decreases as the C: A ratio increases, and this decline was directly proportional to the reduction in CS and can be attributed to the aspects such as the weakening effect, the presence of lightweight aggregate, microvoids, and a reduced active phase reaction in higher mix ratios. These factors weaken the bond between the binder and aggregate due to the lower cement content.

Microstructure

The SEM micrographs of SMB manufactured for various C: A mix is shown in Fig. 17. From micrographs analysis, it can be observed that dense and compact microstructure of blocks decreased as the C: A ratio increased, which involved a decreased quantity of OPC generating less amount of cement hydrated product, i.e., C-S-H gel which, in turn, affect the strength of the concrete product. Figure 17(a) indicates a micrograph of SMB made with a C: A ratio of 1:6 exhibiting a dense microstructure with fewer pores, cracks, and unreacted particles. In comparison, other mix ratios 1:11 and 1:15 shown in Fig. 17(b) and (c) exhibited microstructure with increased tiny pores and some unreacted cement particles, which was attributed to mixes containing a reduced quantity of binder (OPC) than that in the mix of 1:6. In Fig. 17(d) showing micrograph of mix ratio 1:21 it can be observed that presence of macrocrack as well as unreacted OPC particles reducing the overall performance of the concrete product. In addition, a higher amount of loose particles and the shrinking of the cement paste could have contributed to the development of macrocracks. A similar pattern of results can be seen in previous research done by Pengfei Ren et al. [67] and Rahul Singh et al. [21]. Therefore, from all the micrographs, it can be noted that the increased C: A ratio results in inferior microstructure properties, decreased CS, and increased water absorption.

Carbon emission and cost analysis

Historically, carbon emissions are the overall greenhouse gas emissions generated by an organization, event, product, or individual. Accurately calculating the total carbon footprint is impractical due to the data required and the natural occurrences that produce CO_2. Since carbon dioxide is the primary greenhouse gas emitted during the production of binder materials, typically from the combustion of fossil fuels, our analysis focused solely on CO_2 emissions. The study included calculating CO_2 emissions (CO_2-e) for specific components of concrete mix ratios based on one cubic meter of concrete volume.

Carbon emissions and cost-effectiveness in SMB were assessed using the engineering environmental index (EEI) and engineering economic index (EEI). These indices were calculated by proportionally relating structural efficiency to carbon emissions and cost, following methodologies from previous studies by Nayaka et al. [66] and Kanadasan and Razak [68]. The CO_2 emissions from the production of each concrete mix ingredient were determined using data from

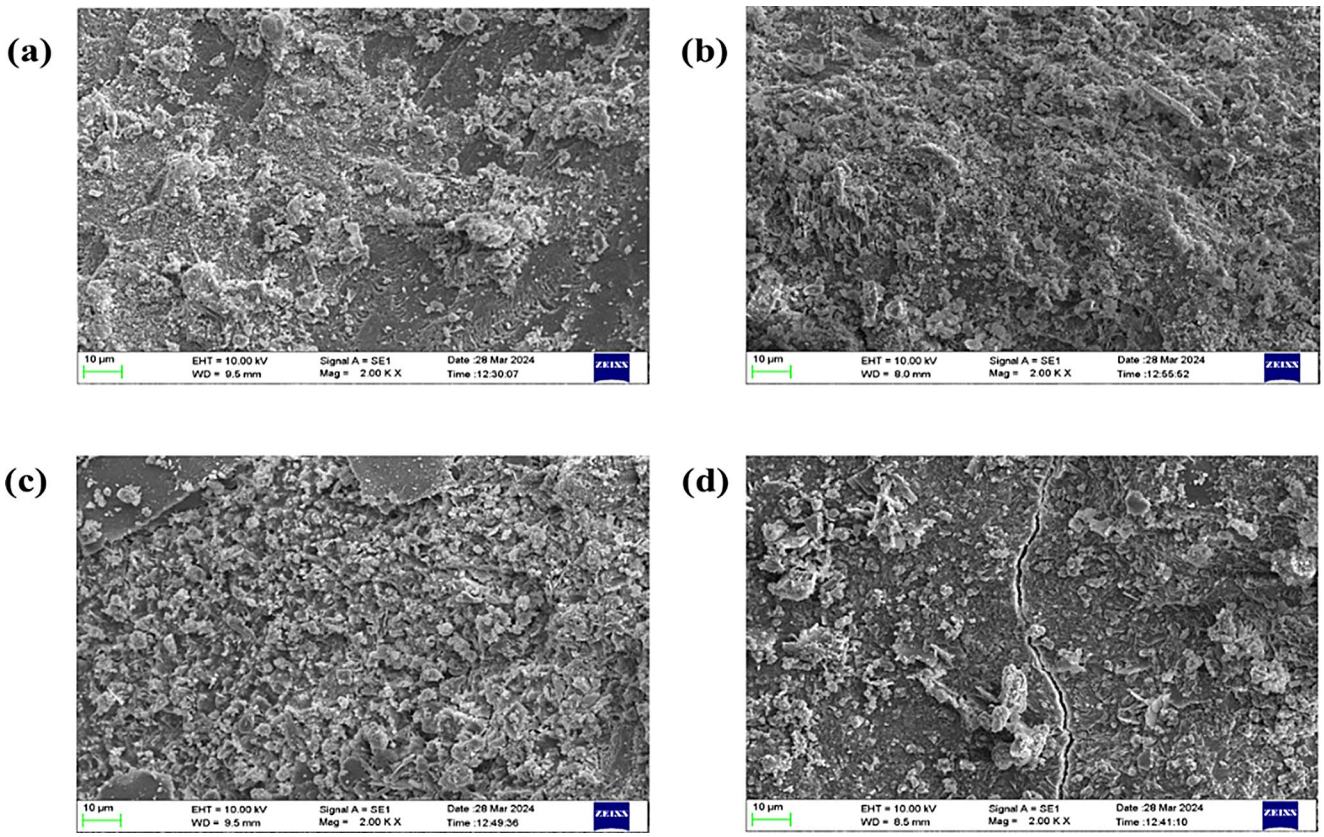


Fig. 17 SEM of SMB made of mix ratio (**a**) 1:6 (**b**) 1:11 (**c**) 1:15 (**d**) 1:21

Table 4 Total CO_2 emissions, CO_2 reduction, and engineering environmental index (EEI) for different mix ratios

Mix Ratio (C: A)	OPC CO_2-e (t/m^3)	Water CO_2-e (t/m^3)	FRCA CO_2-e (t/m^3)	CRCA CO_2-e (t/m^3)	Total emission CO_2-e (t/m^3)	Reduction in CO_2 (%)	EEI (MPa/(kg/m^3)) / (t CO_2-e/m^3)
1:6	0.2479	0.0000326	0.0087	0.0223	0.2789	-	0.0401
1:8	0.1985	0.0000262	0.0093	0.0238	0.2315	16.97	0.0425
1:11	0.1531	0.0000200	0.0098	0.0252	0.1881	32.54	0.0452
1:13	0.1328	0.0000175	0.0101	0.0258	0.1687	39.50	0.0475
1:15	0.1175	0.0000153	0.0103	0.0263	0.1540	44.78	0.0460
1:18	0.0996	0.0000130	0.0105	0.0268	0.1369	50.91	0.0413
1:21	0.0867	0.0000113	0.0106	0.0272	0.1245	55.36	0.0352
1:24	0.0770	0.0000100	0.0107	0.0275	0.1152	58.69	0.0167

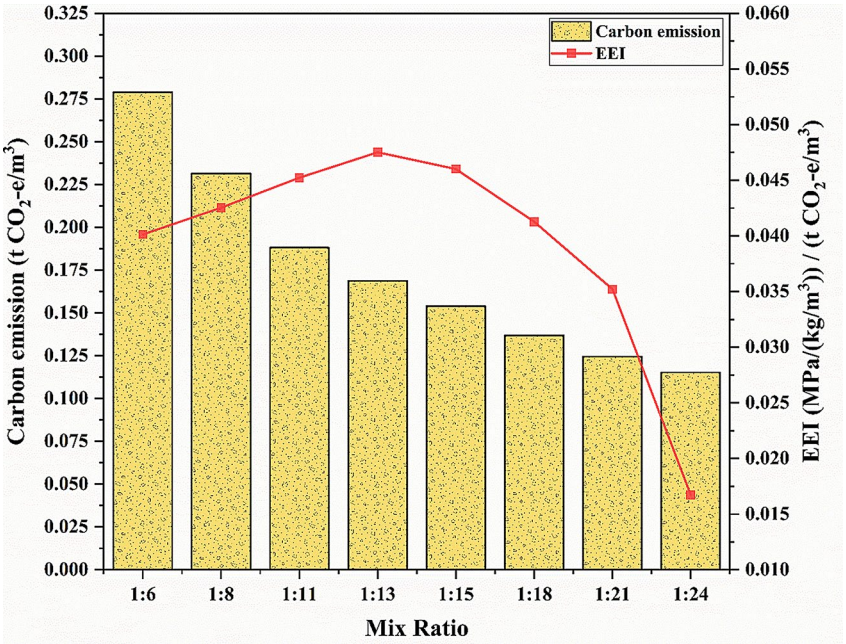


Fig. 18 Relationship between the carbon emission and engineering environmental index for different mix ratios

existing literature. This research used the embodied carbon gate-to-gate method to quantify the carbon emission of all the mixed portions. According to earlier research, during OPC clinker's calcination and production stages, an average of 0.81 kg of CO_2 is produced per kilogram of cement, equivalent to 0.81 t CO_2-e/t [69]. For RCAs, analytical findings done by P. Plaza et al. indicate that FRCA production generates 0.012 t CO_2-e/t, while CRCA production results in 0.02 t CO_2-e/t [70]. Based on the literature, water's carbon emissions are approximately 0.000213 t CO_2-e/t [71]. Table 4 shows each mix ratio's total CO_2 emissions, CO_2 reduction, and engineering environmental index. Figure 18 illustrates the relationship between carbon emissions and the engineering environmental index for various mix ratios. It was noted that a higher C: A mix ratio showed lesser CO_2 emissions compared to a lower C: A mix ratio. For instance, for the permissible CS according to IS 2185 (Part 1)-2005 [46], a C: A ratio of 1:21 mix achieved a 55.36% reduction in CO_2 emissions compared to a 1:6 mix. Similarly, as per ASTM C90 [53], the 1:13 mix achieved a 39.50% reduction in CO_2 emissions compared to 1:6.

This study also emphasizes the additional benefits of using RCAs beyond lowering carbon emissions. Firstly, replacing natural sand with fine recycled aggregates helps reduce erosion and landscape damage, reducing global environmental issues like climate change. Secondly, repurposing demolished concrete conserves natural resources, reduces waste, and eases the burden on landfills. Finally, utilizing recycled materials diminishes the need for land conversion, helping to protect natural habitats and reduce land usage.

Using RCAs also saves costs [72]. RCAs are waste materials with no inherent economic value, so their cost is minimal or negligible compared to NAs. The procurement costs for 1 kg of each raw material are outlined in Table 5. Since precise cost values for RCAs are not available due to their direct procurement from demolition sites and subsequent processing, the costs of RCA materials are typically referenced from various literature sources. The recycling

Table 5 Cost of materials

Materials	Local material cost (INR/kg)	Material cost (USD/kg)	References
OPC	8.30	0.0922	[70]
FRCA	0.7452	0.0083	[75]
CRCA	1	0.0111	[74]
Water	0.03	0.0003	[70]

Table 6 Reduction of cost and EEI of SMB for all the mix ratios

Mix Ratio (C: A)	Total cost [INR /m^3 (USD/m^3)]	Reduction in cost (%)	EEI (MPa/(kg/m^3)) / (Rupees/m^3)
1:6	4198.66 (46.65)	-	2.67×10^{-6}
1:8	3802.97 (42.26)	9.42	2.59×10^{-6}
1:11	3440.84 (38.24)	18.05	2.47×10^{-6}
1:13	3278.88 (36.44)	21.91	2.45×10^{-6}
1:15	3155.81 (35.08)	24.84	2.25×10^{-6}
1:18	3012.54 (33.49)	28.25	1.88×10^{-6}
1:21	2909.19 (32.34)	30.71	1.51×10^{-6}
1:24	2831.86 (31.48)	32.55	6.80×10^{-7}

Note: – 1USD = 90 INR

plant should be within 30 km of the demolition site and the final use location to reduce environmental impacts and minimize the cost of RCA materials. Producing 1 ton of RCAs costs approximately one INR per kilogram, 33% and 28% cheaper than river sand and manufactured sand, respectively [73]. The cost factor of materials by weight is determined based on current market rates, as shown in Table 6. This study does not account for transportation, electricity, or labour costs. Contemporary trends emphasize sustainability by replacing conventional materials like NA with RCA, promoting it as an environmentally friendly alternative supporting cost savings and green product use. Implementing RCAs can reduce landfill usage and mitigate environmental hazards. Figure 19 illustrates the relationships between concrete costs for various mix ratios and the engineering economic index.

Conclusions

Laboratory-scale experiments were conducted to evaluate the feasibility of using suitable combinations of recycled concrete aggregates to produce load-bearing wall solid masonry blocks. From the results of the experiments, the following conclusions can be drawn:

1. The hardened density of solid masonry blocks decreased as recycled concrete aggregate content increased. For all mix ratios ranging from 1:6 to 1:24, the density decreased from 2215.23 kg/m³ to 1994.59 kg/m³, with percentage reduction varying between 0.8% and 2.1% for each mix compared to the previous ratio. This trend indicated that recycled concrete aggregates have a lower density, and the increased voids or porosity within the blocks contributes to this reduction.
2. The water absorption of solid masonry blocks increased with increased recycled concrete aggregate content, which ranges between 1.42% and 3.57% for mix ratios from 1:6 to 1:24. The percentage increase in water absorption for each cement-to-aggregate ratio mix varied between 5.5% and 29.2% compared to the previous ratio. This is attributed to recycled concrete aggregate's higher water absorption capacity due to its porous nature and adhered mortar. Despite this increase, all

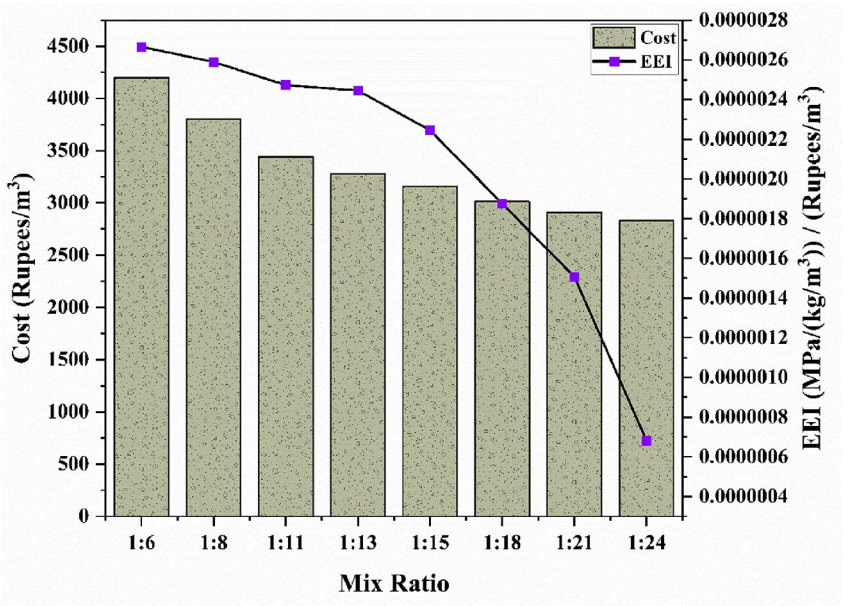


Fig. 19 Relationship between the cost and EEI for different mix ratios

water absorption values met the requirements of the Indian standard code.

3. The compressive strength of Solid masonry blocks decreased as the cement-to-aggregate ratios increased from 1:6 to 1:24 in dry and wet conditions; the values ranged from 24.79 MPa to 2.84 MPa. The compressive strength of SMBs in each cement-to-aggregate ratio mix from 1:6 to 1:15 compared to the previous ratio has decreased, varying between 7.4% and 17.3%; similarly, for cement-to-aggregate ratio mix from 1:18 to 1:24, it has decreased between 21.5% and 58.3%. However, the mix ratios from 1:6 to 1:21 met the requirements specified by the Indian standard code.
4. In both dry and wet conditions, the flexural strength of solid masonry blocks decreased as the cement-to-aggregate ratios increased from 1:6 to 1:24, ranging from 4.21 MPa to 0.61 MPa, with percentage reductions varying between 6.3% and 32.5% for each mix compared to the previous ratio. Despite this decrease, all mixed ratios met the requirements specified by the Indian standard code.
5. The results collectively indicated the practicality of utilizing recycled concrete aggregates from construction and demolition waste as coarse and fine aggregates in manufacturing load-bearing wall blocks. It can be recommended that a cement-to-aggregate ratio of up to 1:21 in solid masonry blocks is suitable for use as a building material in the construction of structural load-bearing walls.
6. The reduced carbon emissions and costs associated with these recycled concrete aggregate solid masonry blocks across various ratios signify a shift towards cost-efficient and eco-friendly materials for affordable housing solutions.

Future research study

The following future research objectives were determined from the research outline.

1. Investigate the long-term durability performance of RCA masonry blocks when subjected to different environmental factors, including freeze-thaw, wet-dry cycle, and chemical exposure.
2. Determine the long-term creep and shrinkage characteristics of RCA masonry blocks, evaluating the impact of recycled aggregates on these properties as time progresses.
3. Evaluate RCA masonry blocks' long-term thermal performance and insulation properties, especially under varying climatic conditions.
4. Investigate how RCA masonry blocks perform under cyclic loading over extended durations, examining any potential deterioration in structural integrity.

Funding Open access funding provided by Manipal Academy of Higher Education, Manipal. The authors declare that no funds, grants, or other support were received during the preparation of the manuscript.

Data availability All data available in the study.

Declarations

Ethical approval This paper neither was published nor is under review elsewhere.

Informed consent All the authors are aware of this paper.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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